Recommendations Report: Predicting First-Deposit Activation

To: STADIOEquities Management
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Subject: Model selection, improvement and deployment recommendations

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Executive summary

Two models were built to predict customer conversion, using the public UCI Bank Marketing dataset as a proxy for first-deposit activation. Logistic regression is recommended, subject to recalibration, cost-based thresholds and the use of the behavioural data requested in SS1.

1. Which model is best?

Model 1 (logistic regression) is recommended over Model 2 (decision tree). Both models were trained on earlier campaigns and tested on the most recent 20% of customers, reflecting how STADIOEquities would learn from past registrants to predict future ones.

Metric (test period)	Model 1: Logistic Regression	Model 2: Decision Tree
ROC AUC (95% CI)	0.741 (0.728 to 0.752)	0.552 (0.539 to 0.567)
PR AUC	0.539	0.389
Recall	0.932	0.693
Precision	0.352	0.306
F1-Score	0.511	0.424
Brier score (lower is better)	0.315	0.266

Table 1: Test-period performance of both models.

Model 1 ranks customers far better and outperforms Model 2 on most classification measures. It identified 93.2% of converting customers, compared with 69.3% for Model 2. Paired bootstrap tests showed all of these differences to be statistically significant ($p < 0.001$), and McNemar's test confirmed that Model 1 makes significantly fewer classification errors. In practical terms, if STADIOEquities could only follow up the top 20% of new registrants, Model 1 would reach 38.5% of eventual converters against 31.4% for Model 2.

The most important insight is why Model 1 wins. On the training period and on a random data split, the two models were statistically indistinguishable (cross-validation AUC 0.671 versus 0.665, $p = 0.373$; random-split AUC 0.800 versus 0.799). The gap appeared only when economic conditions in the test period moved outside the range seen in training. A decision tree cannot extrapolate beyond the values it has learned, so its AUC fell by 0.247, while logistic regression fell by only 0.059. Because STADIOEquities will operate in changing market conditions, robustness to change should be the deciding criterion. Logistic regression also produces coefficients that can be explained to regulators and customers.

2. How can the selected model be improved?

Four improvements are recommended.

First, recalibrate the probabilities. Class weighting, used to handle the class imbalance, caused Model 1 to overestimate conversion probabilities: customers scored at about 0.6 converted only about 25% of the time, and Model 2 achieved a better Brier score (0.266 versus 0.315). Platt scaling or isotonic regression, fitted on a separate validation period, would correct this without affecting the model's ranking ability.

Second, set the decision threshold using business costs rather than the default of 0.5. At 0.5, Model 1 labelled 81.6% of customers as likely converters. The threshold analysis showed that at 0.6, F1 improves from 0.511 to 0.570 while 82.6% of converters are still identified. The final threshold should be selected on a validation period, weighing the cost of an onboarding intervention against the value of an activated account.

Third, retrain on a rolling window and monitor drift. The conversion rate rose from 6.4% in the training period to 30.8% in the test period, showing that customer behaviour can shift substantially. Monthly retraining, combined with monitoring of input distributions and conversion rates, would keep the model current (Moro, Cortez and Rita, 2014).

Fourth, reduce reliance on period-specific features. Both models depended heavily on month of contact and highly correlated economic indicators, which describe when a customer was contacted rather than who the customer is. Stronger regularisation or removal of these features would improve stability. A tree ensemble such as a random forest could also be tested, as Sifa et al. (2015) found random forests strongest, but it should be adopted only if it outperforms logistic regression under time-ordered evaluation.

3. How must the model be adapted for the SS1 data?

The modelling pipeline can be reused, but its target, features and validation must change. The target becomes whether a customer makes a first deposit within 30 days of registration, predicted at day 3 after registration. Only information available at day 3 may be used, applying the same leakage principle that led to the removal of call duration in this study. Only registrants with a full 30 days of follow-up should be included in training.

The campaign and economic variables should be replaced with the fields requested in SS1: registration attributes (such as acquisition channel, device and age band) and behaviour during the first three days, including number and length of sessions, time to first login, completion time for identity verification, visits to the funding page and deposit attempts, together with the timing of this activity. These would be engineered into counts and rates within the 72-hour window and passed through the existing preprocessing pipeline. The time-ordered split should be based on registration date, class weighting should be retained, and personal information should be handled in line with the Protection of Personal Information Act (POPIA).

4. Do the results align with the literature from Part A?

The results largely align with the literature, with important differences in setting. Consistent with Moro, Cortez and Rita (2014), customer and campaign information predicted conversion,

economic indicators such as the Euribor rate were highly influential, and time-aware evaluation proved essential. However, their study used a larger engineered feature set and more complex models, so its results are not directly comparable with the two models tested here.

Sifa et al. (2015), whose new-user setting is closest to STADIOEquities, handled a very small buying class (under 2%) with oversampling. This was supported here: class weighting enabled Model 1 to identify 93.2% of converters despite only 6.4% of training customers converting. Their study favoured random forests under ten-fold cross-validation, whereas the decision tree here performed well only on a random split. This suggests that random splits can overstate the reliability of tree-based models when conditions change, and that the difference in setting (game purchases rather than bank deposits) also matters.

Sakar et al. (2019) found that combining click-stream and session-level features improved purchase prediction for visitors with no history. This study supports that direction: with mainly static campaign-level features, discrimination remained moderate, which strengthens the case for the session and timing data requested in SS1. Their data, however, came from online retail with real-time behaviour.

Conclusion

Logistic regression should be adopted as the STADIOEquities activation model. With recalibration, cost-based thresholds, regular retraining and day-3 behavioural data, it offers a robust and explainable basis for prioritising onboarding support for new registrants.

References

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